# When Observation Choices Resemble Dynamics in AI Conversation Archives

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**Abstract.** We revisit conversation-network and embedding measurements through their observation process. Fixed-graph revelation has mechanical growth expectations, while matched forecast origins change a user–assistant history contrast on exposed PRISM data. These case studies support explicit reporting of observation units, available information and controls before interpreting structure as cognitive dynamics.

***Keywords.*** *Conversation networks; Semantic embeddings; Temporal nulls; Measurement; Human–AI interaction*

## 1 From archive maps to claims about development

Conversation archives support useful representations of human–AI interaction, but a representation's evolution need not identify the process that generated the interaction. The published Cognitive MRI case study constructed a semantic similarity network and described communities and bridge conversations [6]. The accepted ISCS 2026 hierarchy study related episodes to extracted concepts and interpretably chosen abstraction levels [7]. Both draw on the same 1,908-conversation author archive; they are not independent replications. Their descriptive constructions motivate, but do not validate, measurements of memory, learning or intellectual influence.

We synthesize two completed calibration studies around one question: *which apparent dynamics depend on how an archive becomes observable?* The first concerns revealing vertices of a fixed endpoint graph; the second concerns which messages are available when forecasting the next contribution. The contribution is an auditable methodological case study linking network and sequence measurements, with explicit limits on cognitive interpretation. No new conversations or confirmation sample are introduced here.

## 2 Revelation of a fixed graph already produces growth

Let $G$ be a fixed simple graph on $N \geq 2$ conversation vertices with $M$ edges and degrees $d_i$. Whole-conversation embeddings and a fixed similarity threshold determine its edges. Reveal vertices in a uniformly random order and retain the induced graph after $k$ arrivals. Its edge count $M_k$ satisfies

$$\mathbb{E}[M_k] = M\frac{k(k-1)}{N(N-1)}. \tag{1}$$

Each endpoint pair is included with the same probability, so linearity of expectation gives this result for every endpoint graph. Clustering is not required. A growing edge curve alone therefore cannot distinguish semantic development from graph revelation.

The vertex denominator also matters. All arrivals number $k$, whereas active vertices $A_k$ exclude isolates. For $1 \le k \le N$,

$$\mathbb{E}[A_k] = \sum_{i=1}^{N} \frac{k}{N} \left[ 1 - \frac{\binom{N-1-d_i}{k-1}}{\binom{N-1}{k-1}} \right]. \tag{2}$$

The term in brackets is the probability that an included vertex has an included neighbor. We verified both identities by exhaustive subset enumeration of small graphs, including isolates and complete graphs, using the existing calibration implementation. These are expectation identities, not formulas for the expected slope of a fitted log–log regression.

Population-versus-activity distinctions already occur in temporal-network densification research [5]; structured temporal reference models are established [3]. Their application here requires declaring which order information is destroyed. Uniform permutations and block reorderings that retain internal order answer different questions. A residual under either control remains conditional on its exchangeability assumptions; nonrejection does not establish that all observed growth is mechanical. Likewise, placing a completed conversation's features at its creation time is retrospective observation unless feature availability is independently established.

# 3 Forecast origin changes a role contrast

The second case reuses exposed development results from PRISM [4]: 1,191 selected conversations in 360 dependence components, grouped using participant identifiers and duplicate links. These are grouping proxies, not independently verified longitudinal individuals. The probe portion contains 240 conversations in 72 components and 428 paired user/assistant targets. Earlier fit and tuning partitions contain 216 and 72 components. All these data were inspected in previous pilots; the separate 248-component reserve remains unused.

For each target, we compare normalized uniform pooling of available message vectors with pooling the last two messages. Let $g_r$ be the mean difference in target cosine similarity for role $r$, and let $I = g_{\text{assistant}} - g_{\text{user}}$. Message positions and target embeddings are fixed. The native view uses at most the first 128 content tokens per message, with targets after at least two completed exchanges and within the first twelve messages. MiniLM-L6-v2 and E5-small-v2 are two instruments on the same data.

The *natural* origin includes every message before each target: assistant prediction sees the current user request. The *common-prefix* origin predicts both members of a pair from the same preceding completed exchange, withholding that request. User predictions are unchanged. Assistant prediction now has a longer forecast horizon, so this comparison does not isolate a causal effect of human versus artificial memory.

Under natural origins, $I = -0.0869$ for MiniLM and $-0.0215$ for E5. Under the common prefix, the values become $+0.0168$ and $+0.0006$ (Fig. 1). The ordering reverses for MiniLM; E5 is near zero with an interval spanning zero. We do not claim every role difference vanishes. Meanwhile, next-user mean-history gains remain positive: $0.0344$ $[0.0225, 0.0488]$ for MiniLM and $0.0192$ $[0.0160, 0.0226]$ for E5. The earlier fixed checks also retain positive user gains with expanded text, restored message weighting after chunking, and exposure to valid past response candidates. The case constrains a role interpretation while preserving a useful predictive measurement.

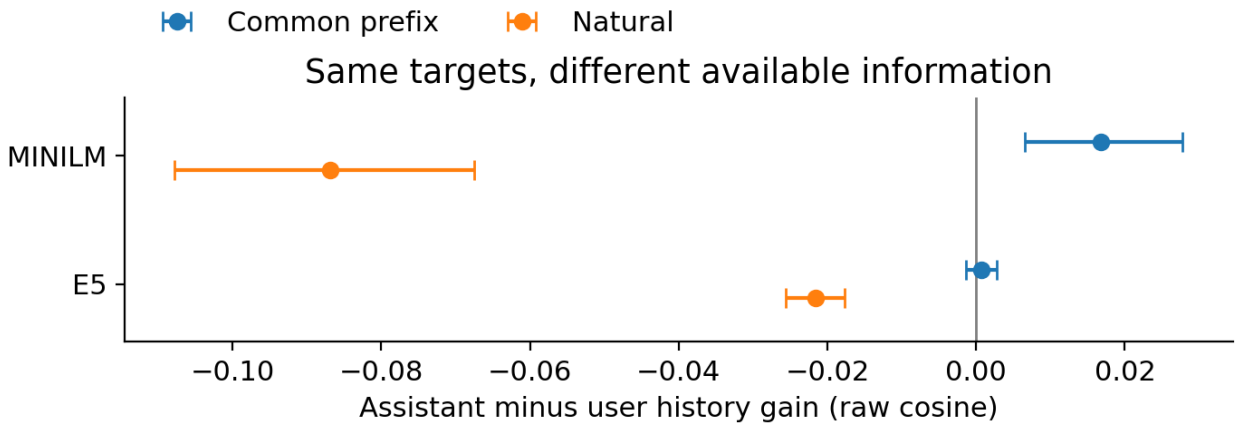



Figure 1: Role interaction $I$ in the native input view. Bars are descriptive 95% component-bootstrap intervals (2,000 draws), conditional on the exposed development sample. Positive values mean a larger mean-versus-last-two gain for assistant targets. Encoder scales need not be comparable. Changing origin withholds information and changes the assistant forecast horizon.

Uncertainty averages targets within conversations and conversations within components, then resamples components with equal weight. It does not treat 428 target pairs as independent people. Synthetic controls from the completed study reproduce a history benefit with a fixed topic and conditionally independent user innovations; a process with planted older-state dependence also produces it. Thus the cosine contrast does not identify a memory mechanism. It is not an accuracy measure for reasoning, learning or useful assistance.

# 4 Contribution boundary and reporting implications

Curved Contrastive Learning already reports dialogue-intent-dependent benefits of full versus recent history [2]. OnePred compares history and intent-memory approaches to next-query prediction [1]. Our synthesis does not claim priority for useful context, role differences or temporal nulls. Its practical contribution is to show, with exact graph identities and a quantified forecast-origin contrast, why the observation process must accompany an interpretation of conversational dynamics.

A report should specify the unit retained, when its features become available, the population denominator, the permitted predictive prefix, and what each control preserves. Graph revelation and forecast withholding are distinct operations; their common lesson is that a changed measurement can reflect a changed observable rather than a changed underlying mechanism. This supports more accountable use of conversation-derived profiles in research and AI memory design, without claiming a demonstrated downstream benefit.

This synthesis combines published descriptive work, established mathematics and exposed development analyses. It establishes neither natural prevalence nor a general theory of memory or semantic dynamics. The evidence package records source versions, all four forecasting views, component effects and replay checks. A stronger empirical paper needs a distinct outcome and independent evidence. This synthesis does not justify reserve evaluation.

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